

Readers Writers

```
readcount = 0
sem_init(&mutex, 1)
sem_init(&writer_or_readers, 1)
sem_init(&service, 1)
```

```
void writer () {
    while(1){
        sem_wait(&service)
        sem_wait(&writer_or_readers)
        sem_signal(&service)
        write(file, data)
        sem_signal(&writer_or_readers)
    }
}
```

```
void reader () {
    while(1){
        sem_wait(&service)
        sem_wait(&mutex)
        readcount += 1;
        if (readcount == 1)
            sem_wait(&writer_or_readers)
        sem_signal(&service)
        sem_signal(&mutex)
        data:=read(file)
        sem_wait(&mutex)
        readcount -= 1;
        if (readcount == 0)
            sem_signal(&writer_or_readers)
        sem_signal(&mutex)
    }
}
```

Dining Philosophers

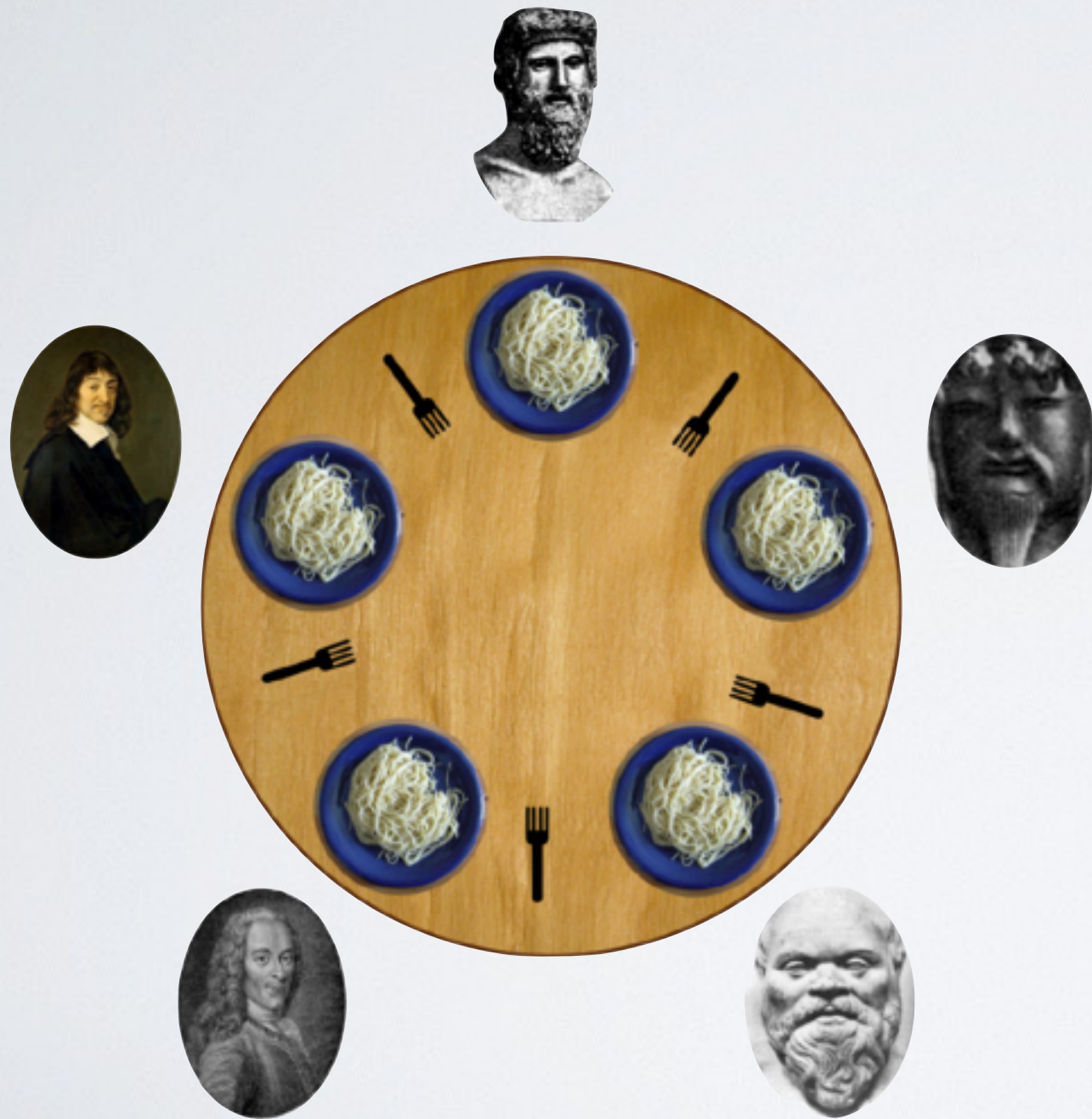


image from wikipedia